

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Crop Data Input	Select State, District, Season, Crop and Area Validate input fields before submission
FR-4	Crop Production Prediction	Fetch historical dataset for selected crop / region Run ML model to predict crop production / yield
FR-5	Result Visualization	Display predicted production in tonnes and yield per hectare Show comparison charts (year-wise / season-wise)
FR-6	Dataset & Model Management	Admin uploads / updates historical crop dataset Admin triggers model retraining and views accuracy metrics
FR-7	Reporting	Generate downloadable report (PDF / Excel) of prediction results View state-wise and crop-wise analytics dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, intuitive interface so that farmers and users with limited technical knowledge can easily input crop details and understand the predicted output.
NFR-2	Security	User data and credentials must be protected using encryption (e.g. HTTPS, password hashing) and role-based access control between users and administrators.
NFR-3	Reliability	The system should consistently return accurate crop production predictions and handle invalid/missing input data gracefully without crashing.
NFR-4	Performance	Prediction results should be returned within a few seconds of submitting the crop details, even with large historical datasets.
NFR-5	Availability	The application should be available with minimal downtime, using cloud hosting so that it can be accessed anytime by farmers and administrators.

FR No.	Non-Functional Requirement	Description
NFR-6	Scalability	The solution should be able to scale to accommodate additional crops, regions, and a growing number of concurrent users without major redesign.